Foreign KEY

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Objectives

- Introduce Foreign Key
- Practical Example of Foreign Key

What is a FOREIGN KEY?

- The FOREIGN KEY constraint is a key used to link two tables together.
- A FOREIGN KEY is a field (or collection of fields) in one table that refers to the PRIMARY KEY in another table.

- Branch Table

branchNo	street	city	postcode
B005	22 Deer Rd	London	SW1 4EH
B007	16 Argyll St	Aberdeen	AB2 3SU
B003	163 Main St	Glasgow	G11 9QX
B004	32 Manse Rd	Bristol	BS99 1NZ
B002	56 Clover Dr	London	NW10 6EU

- Staff Table

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SL21	John	White	Manager	M	1-Oct-45	30000	B005
SG37	Ann	Beech	Assistant	F	10-Nov-60	12000	B003
SG14	David	Ford	Supervisor	M	24-Mar-58	18000	B003
SA9	Mary	Howe	Assistant	F	19-Feb-70	9000	B007
SG5	Susan	Brand	Manager	F	3-Jun-40	24000	B003
SL41	Julie	Lee	Assistant	F	13-Jun-65	9000	B005

- Branch Table

Primary Key

branchNo	street	city	postcode
B005	22 Deer Rd	London	SW1 4EH
B007	16 Argyll St	Aberdeen	AB2 3SU
B003	163 Main St	Glasgow	G11 9QX
B004	32 Manse Rd	Bristol	BS99 1NZ
B002	56 Clover Dr	London	NW10 6EU

- Staff Table

Foreign Key

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SL21	John	White	Manager	M	1-Oct-45	30000	B005
SG37	Ann	Beech	Assistant	F	10-Nov-60	12000	B003
SG14	David	Ford	Supervisor	M	24-Mar-58	18000	B003
SA9	Mary	Howe	Assistant	F	19-Feb-70	9000	B007
SG5	Susan	Brand	Manager	F	3-Jun-40	24000	B003
SL41	Julie	Lee	Assistant	F	13-Jun-65	9000	B005

Creating and Inserting Data to Branch Table

```
CREATE TABLE Branch
(
    branchNo VARCHAR(10) NOT NULL,
    street VARCHAR(20),
    city VARCHAR(15),
    postcode VARCHAR(15),
    PRIMARY KEY(branchNo)
);
```

```
INSERT INTO Branch
(branchNo,street,city,postcode)
VALUES
('B005','22 Deer Rd','London ','SWI 4EH' ),
('B007','16 Argyll St','Aberdeen','AB2 3SU' ),
('B003','163 Main St','Glasgow','Gil 9QX' ),
('B004','32 Manse Rd','Bristol','BS99 INZ' ),
('B002','56 Clover Dr','London','NWIO 6EU' )
;
```

Creating EMP Table

```
CREATE TABLE STAFF
(
    staffNo VARCHAR(6) NOT NULL,
    fName VARCHAR(15),
    lName VARCHAR(15),
    position VARCHAR(15),
    sex CHAR(1),
    DOB DATE,
    salary FLOAT(8,2),
    branchNo VARCHAR(10),
    PRIMARY KEY(staffNo),
    FOREIGN KEY(branchNo) REFERENCES BRANCH(branchNo)
);
```

Inserting Data Into EMP Table

```
INSERT INTO STAFF
(staffNo,fName,lName,position,sex,DOB,salary,branchNo)
VALUES
('SL21','John','White','Manager','M','1945-10-01',30000,'B005'),
('SG37','Ann','Beech','Assistant','F','1960-11-10',12000,'B003'),
('SG14','David','Ford','Supervisor','M','1958-03-24',18000,'B003'),
('SA9','Mary','Howe','Assistant','F','1970-02-19',9000,'B007'),
('SG5','Susan','Brand','Manager','F','1940-06-03',24000,'B003'),
('SL41','Julie','Lee','Assistant','F','1965-06-13',9000,'B005')
;
```

Try the below [Output and Explanation]

```
INSERT INTO STAFF
(staffNo,fName,lName,position,sex,DOB,salary,branchNo)
VALUES
('SL28','Sam','Whitteker','Manager','M','1945-10-31',30000,'B0009');
```

Implement Below Tables with Primary Key and Foreign Key Constraints

Student Table

id	name	address	dob	phone	phone02
1	Saman Perera	123, Colombo	10/12/2002	2345123	
2	Kamal Silva	223, Gampaha	10/31/2005	1203654	55515131
3	Thabitha Jeyaraj	445, Galle	1/21/2008	7123654	12121122
4	Kamal Jeyaraj	223, Jaffna	1/5/2007	1456987	
5	Sam Mohamed	123, Galle	12/4/2006	12554725	

Primary Key = id

Course Table

cid	cname	duration
c01	Foundation	8 Months
c02	Degree	3 Years
c03	Masters	2 Years
c04	PHD	5 Years
c05	Diploma	1 Year

Primary Key = cid

Enrolment Table

cid	studentid	coursefee	dateofPayment
c01	1	250,000	1/5/2023
c02	2	500000	12/31/2022
c03	3	60000	12/4/2022
c04	5	700000	1/31/2023
c01	4	1000000	2/1/2023

Primary key = (cid,studentid)

Foreign Keys = cid(course table), studentid (student table)